

CSE4227 Digital Image Processing

Chapter 10 – Image Segmentation (Part III)

Dr. Kazi A Kalpoma

Professor, Department of CSE

& Director of ICT

Ahsanullah University of Science & Technology (AUST)

Contact: kalpoma@aust.com

Google Class code: cgxyzjp



CSE | AUST

Fall 2020

Today's Contents

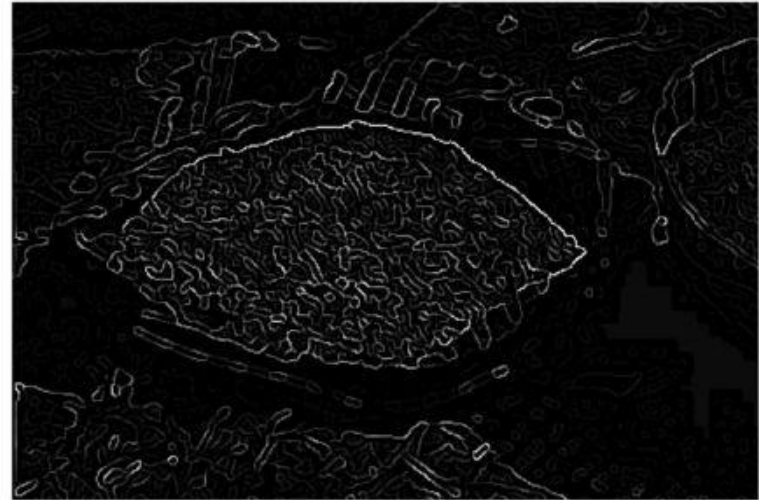
- Introduction to Hough transform
- Using gradient information to detect lines
- Representing a line
 - The $[a,b]$ -representation
 - The $[\rho,\theta]$ -representation
- Algorithms for detection of lines

• Chapter 10 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [**Section 10.2.7**]

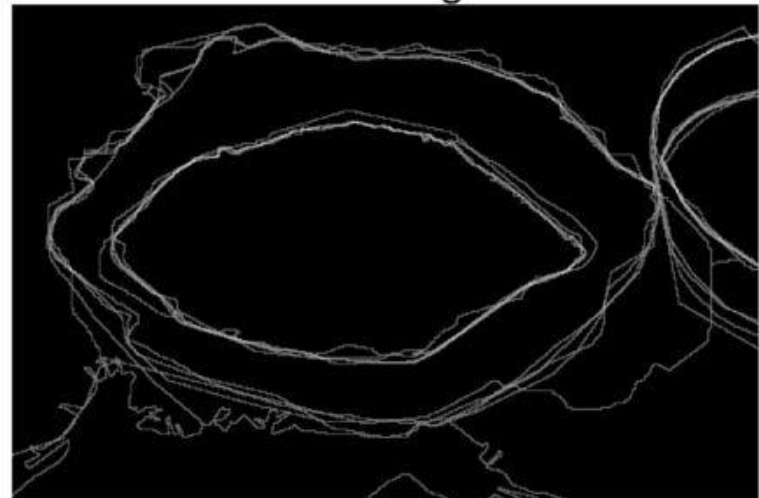
Boundaries of Objects from Edges



Image



Machine Edge Detection

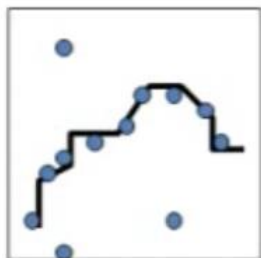


Human Boundary Marking

Knowledge about Boundary

More
↑
Knowledge
↓
Less

- Can extract the complete closed contour of the object if its shape is known
- Can extract pieces of the boundary using general line or curve models
- Can just try to find a connected series of edge elements, using only heuristics on say, edge curvature

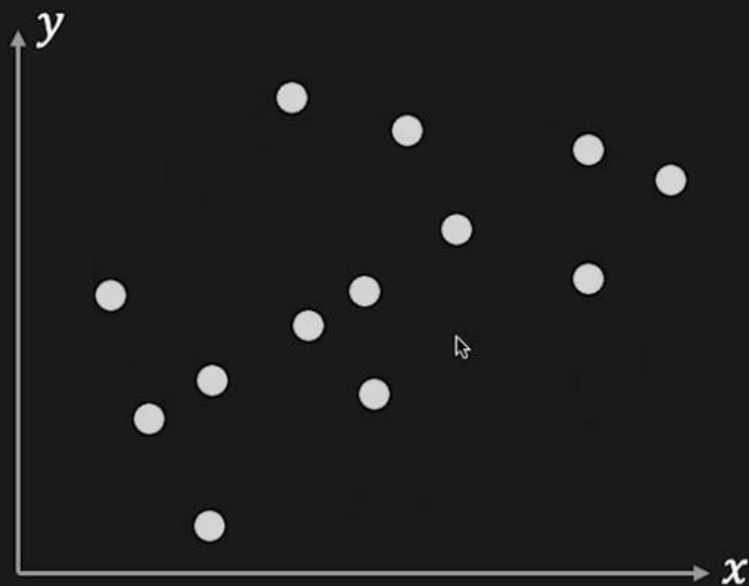


Hough Transform

- ❑ A way of finding edge points in an image that lie along a straight line or curve.
- ❑ Introduced by Paul Hough (1962)
- ❑ First goal is to find the location of lines in image.
- ❑ Given a set of edge points, find the line(s) which best explain the data.
- ❑ Relatively unaffected by gaps in lines, curves, and noise.
- ❑ Hough Transform is used to connect the disjointed edge points.

Hough Transform: Line Detection

Given: Edge Points (x_i, y_i)



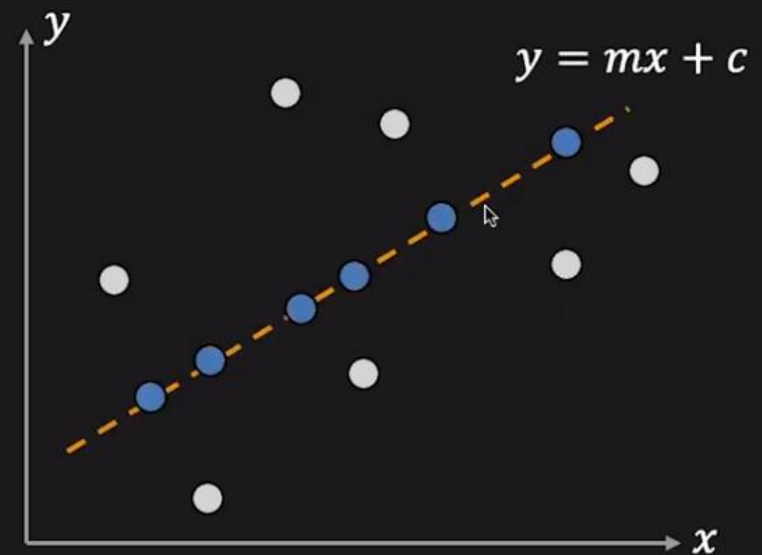
Hough Transform: Line Detection

Given: Edge Points (x_i, y_i)

Task: Detect line
 $y = mx + c$

m = slop

c = intercept of the line



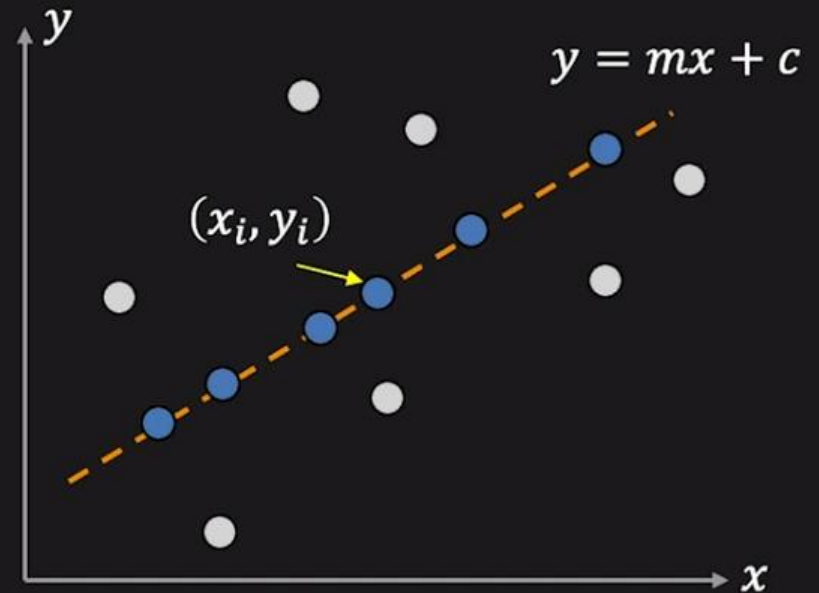
Hough Transform: Line Detection

Given: Edge Points (x_i, y_i)

Task: Detect line
 $y = mx + c$

m = slop

c = intercept of the line



Consider point (x_i, y_i)

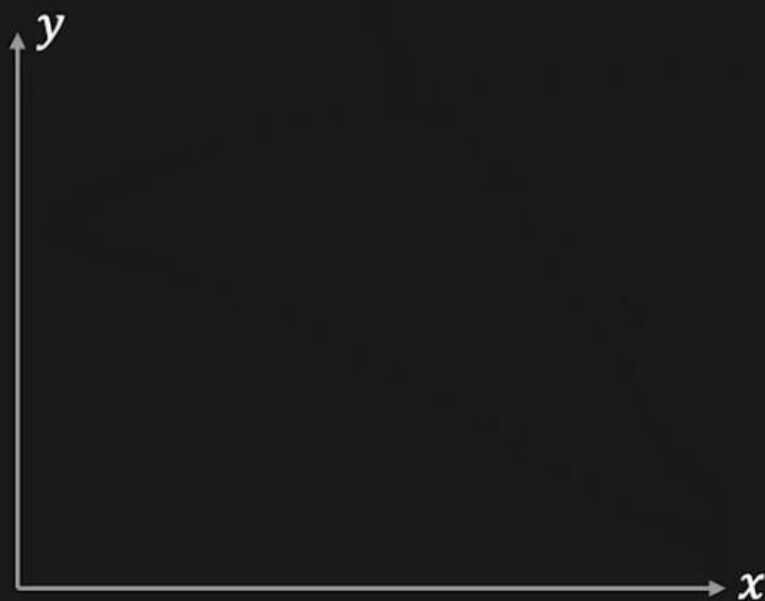
$$y_i = mx_i + c$$



$$c = -mx_i + y_i$$

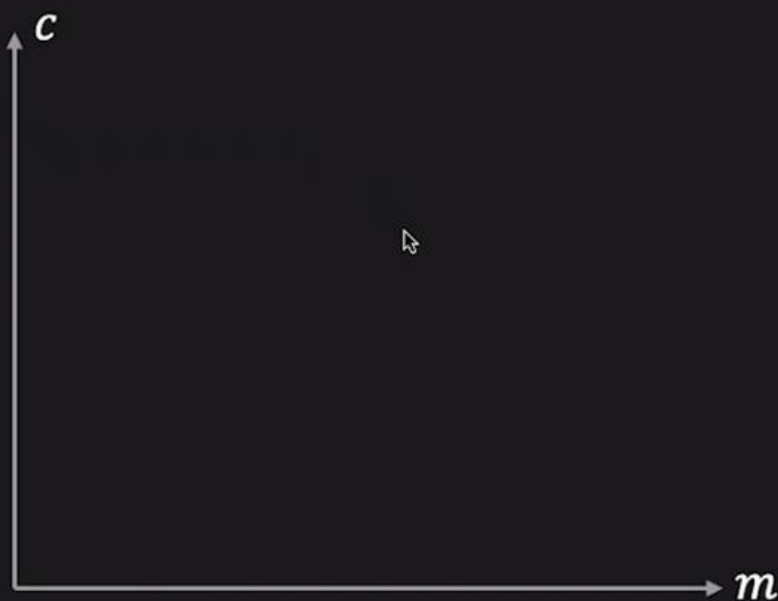
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

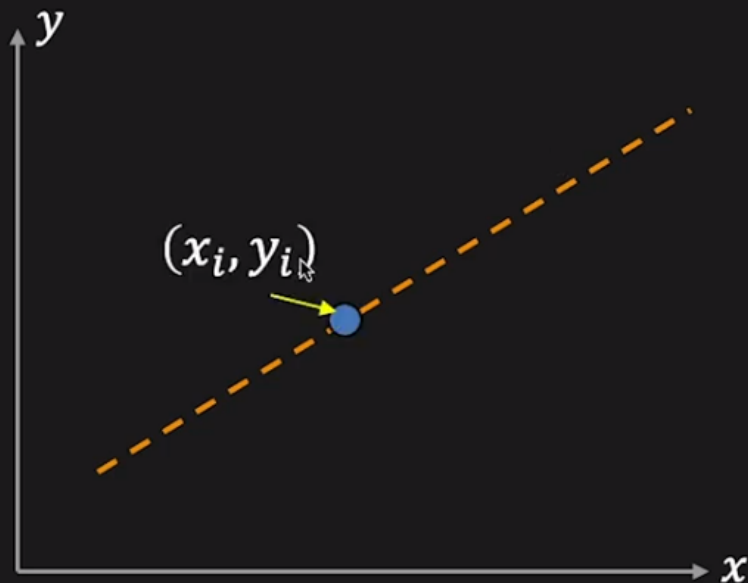
Parameter Space



$$c = -mx_i + y_i$$

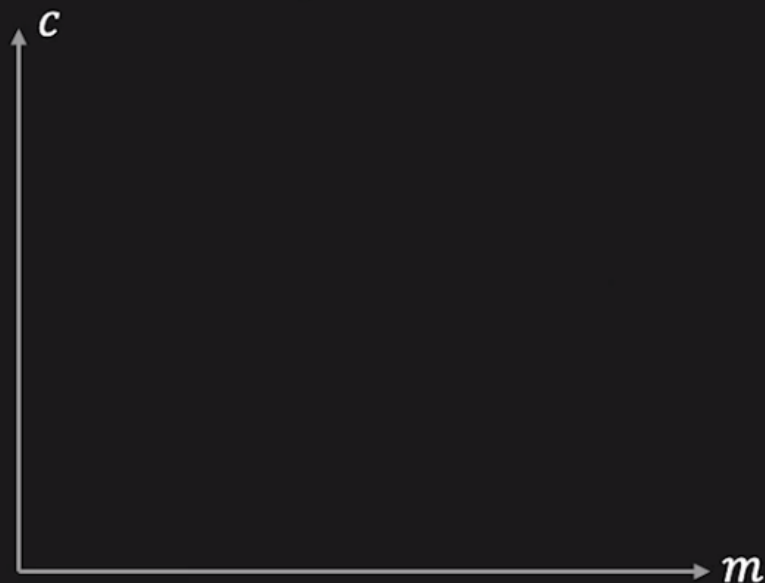
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

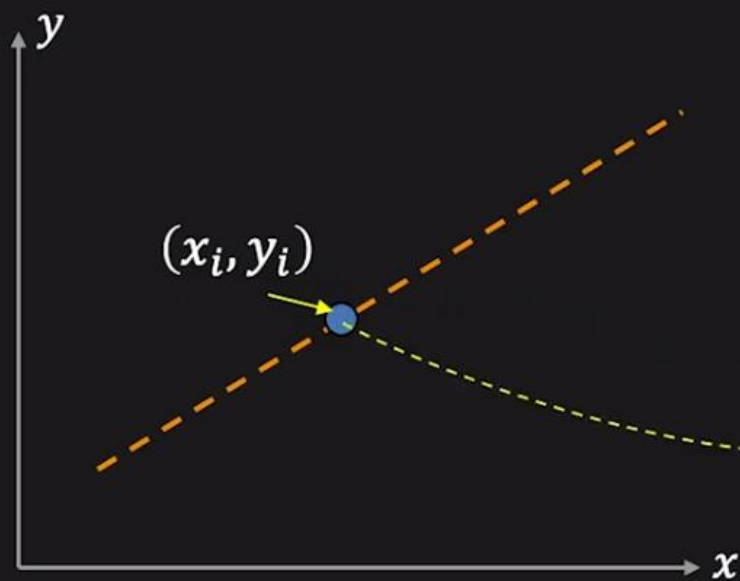
Parameter Space



$$c = -mx_i + y_i$$

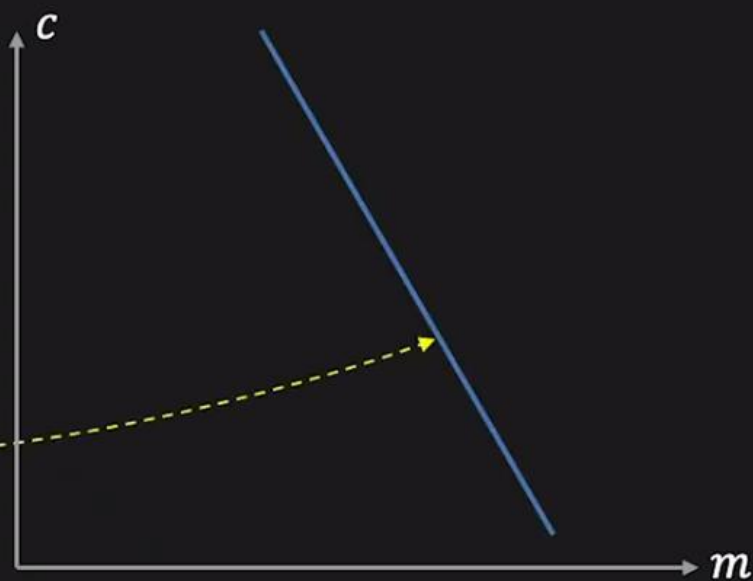
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

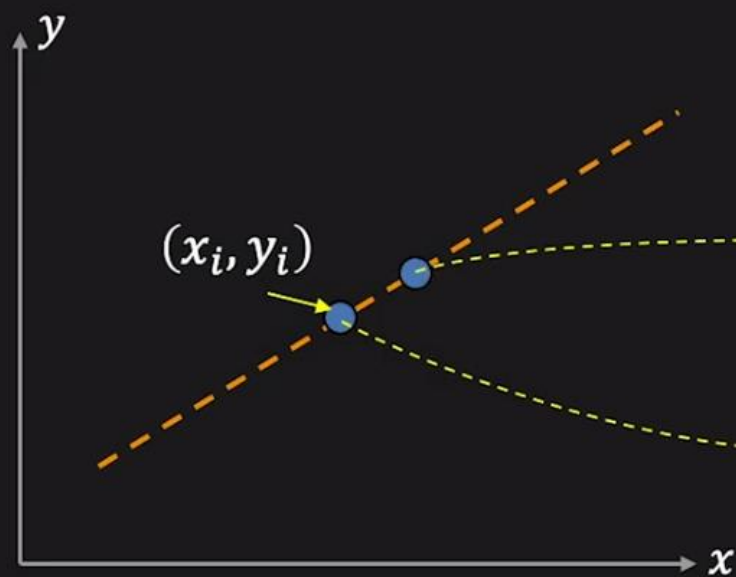
Parameter Space



$$c = -mx_i + y_i$$

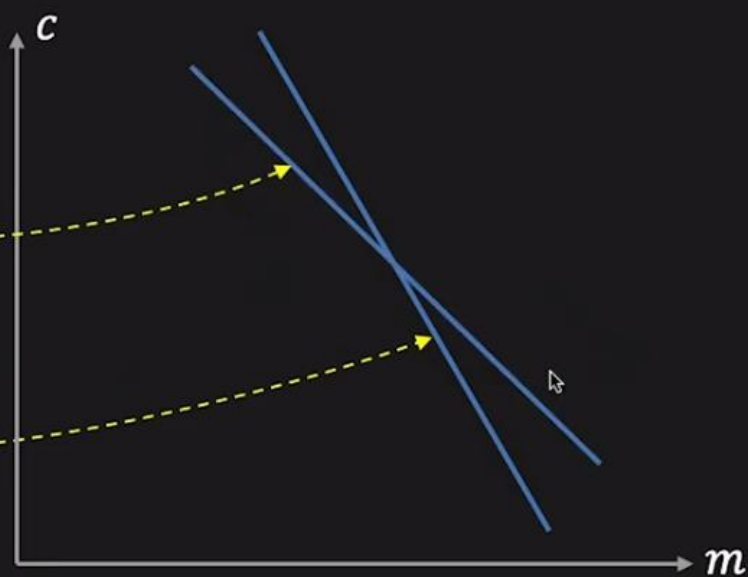
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

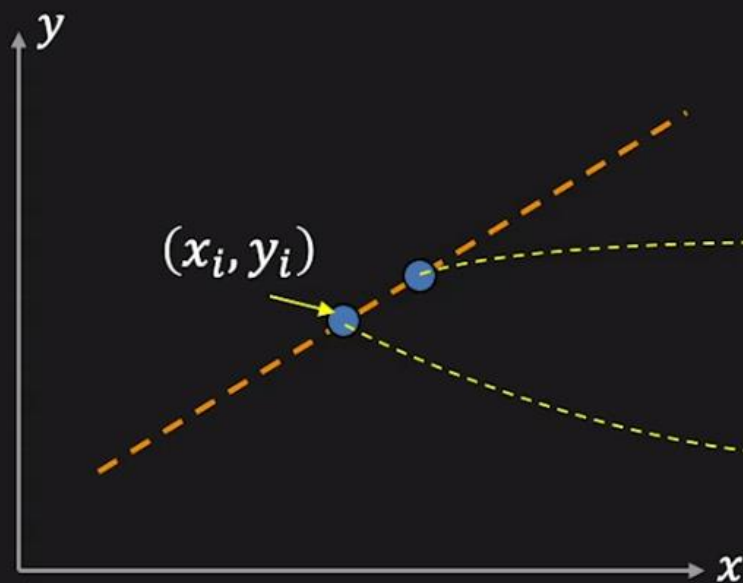
Parameter Space



$$c = -mx_i + y_i$$

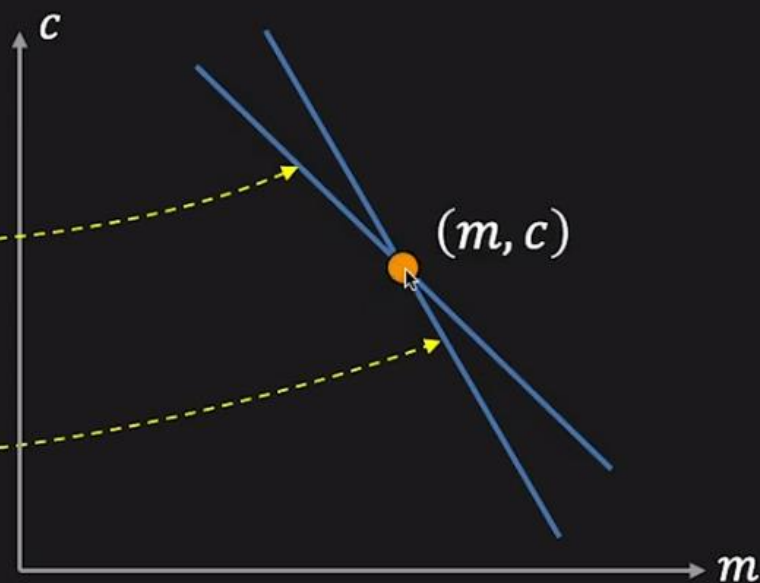
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

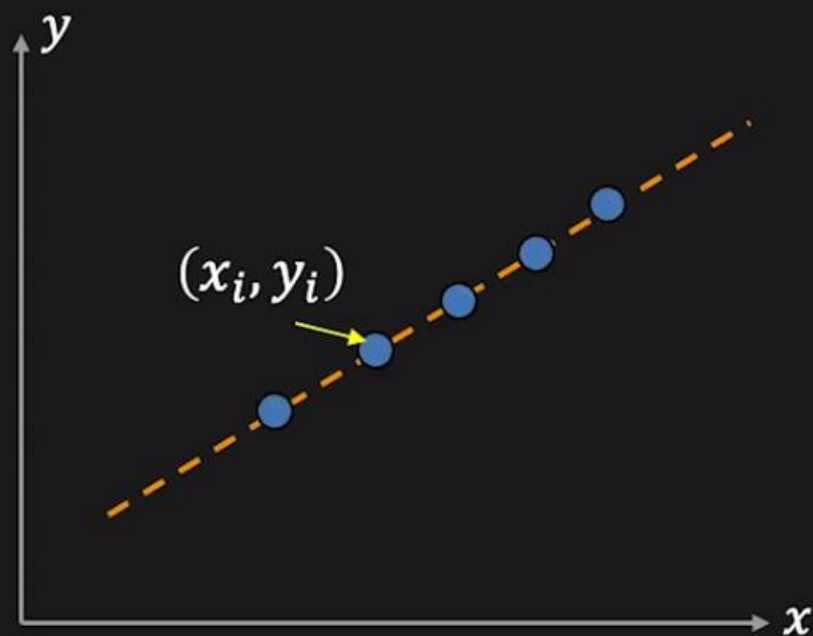
Parameter Space



$$c = -mx_i + y_i$$

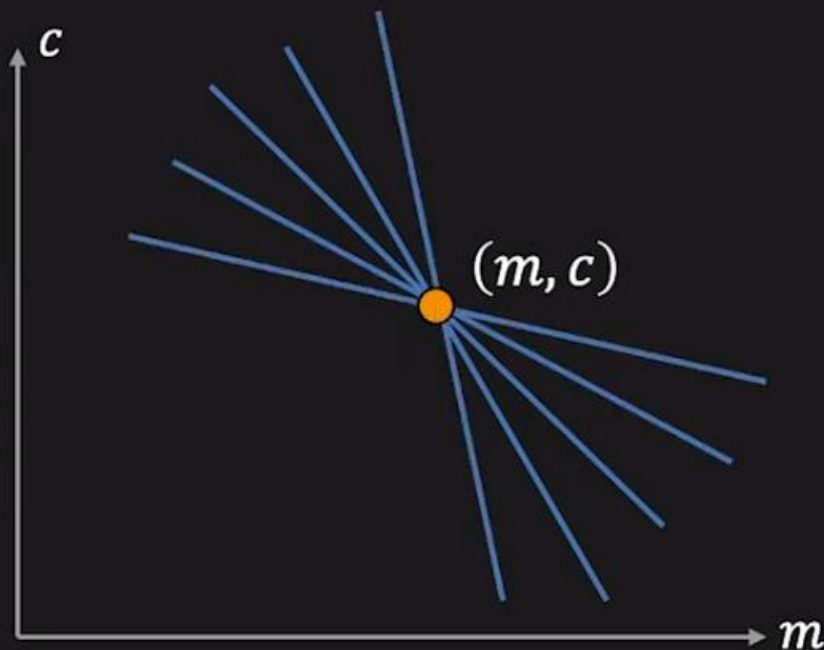
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

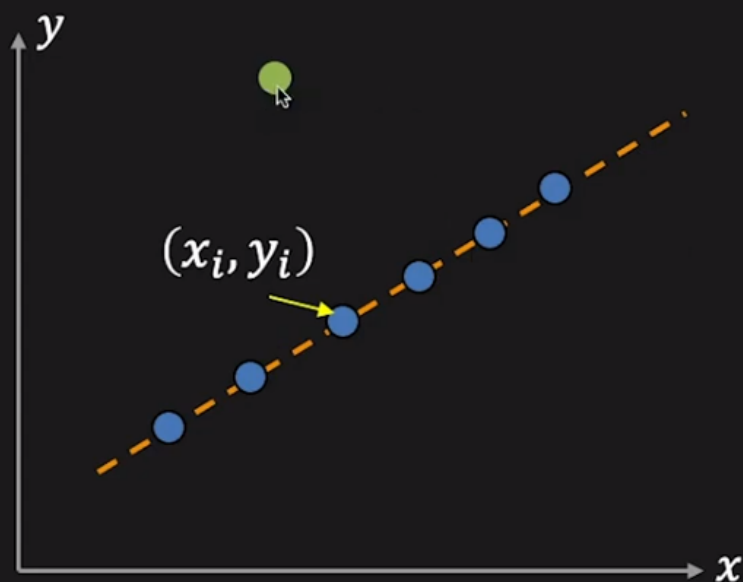
Parameter Space



$$c = -mx_i + y_i$$

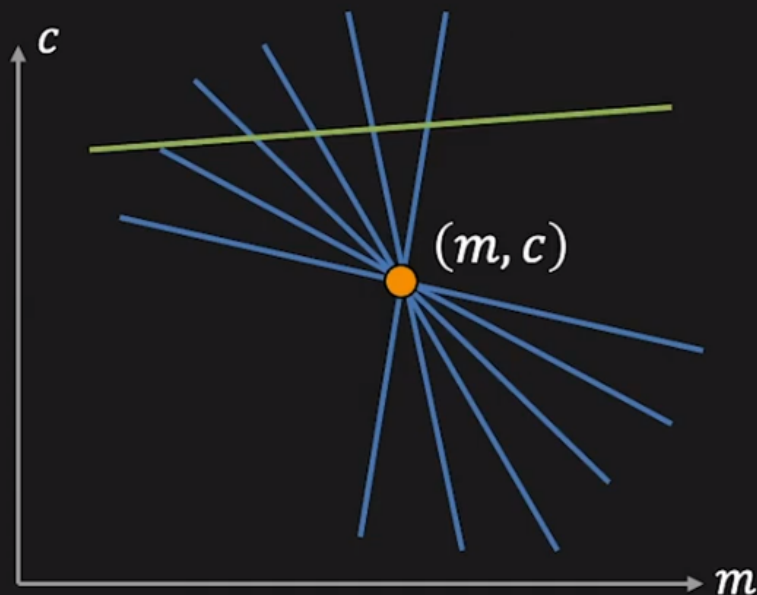
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

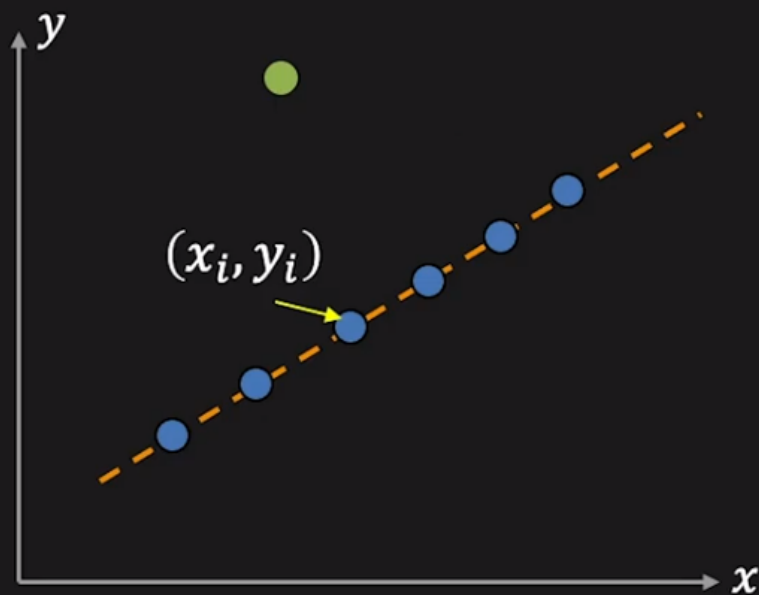
Parameter Space



$$c = -mx_i + y_i$$

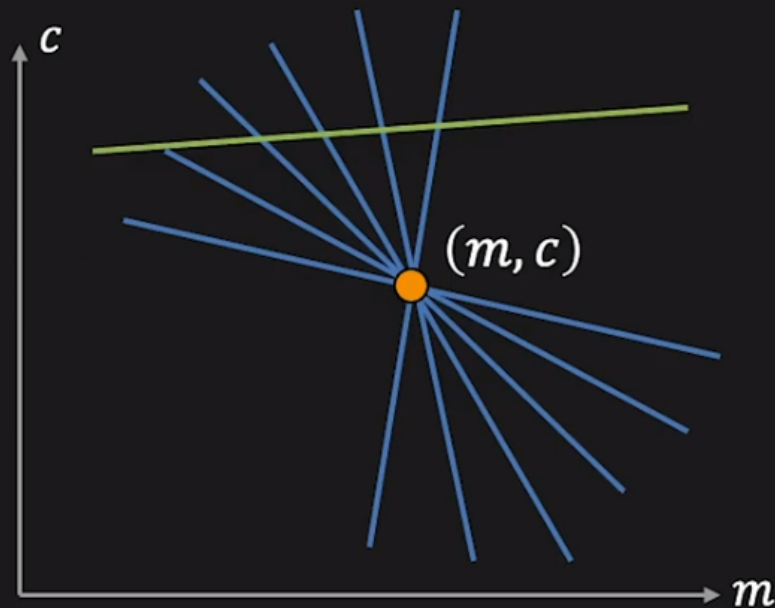
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

Parameter Space



$$c = -mx_i + y_i$$

Point $\longleftrightarrow$ Line

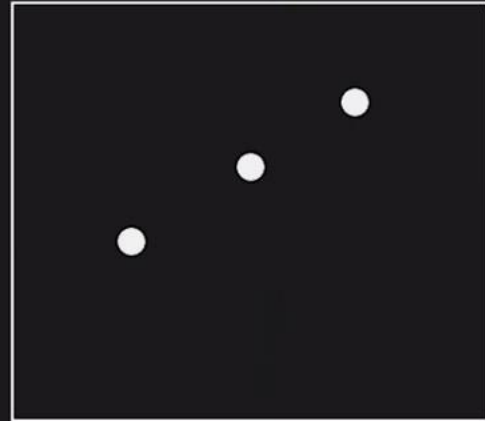
Line $\longleftrightarrow$ Point

Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create **accumulator array** $A(m, c)$

Image



$A(m, c)$



Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create **accumulator array** $A(m, c)$

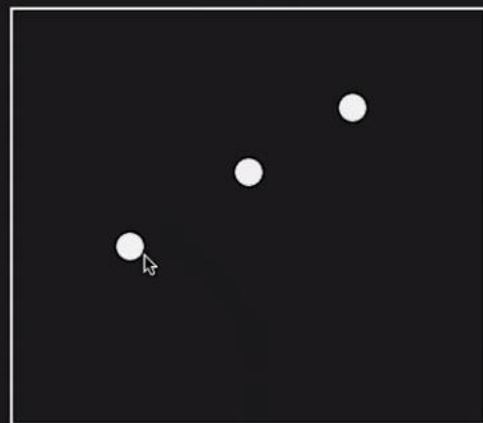
Step 3. Set $A(m, c) = 0$ for all (m, c)

Step 4. For each edge point (x_i, y_i) ,

$$A(m, c) = A(m, c) + 1$$

if (m, c) lies on the line: $c = -mx_i + y_i$

Image



$A(m, c)$

c	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0

Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create **accumulator array** $A(m, c)$

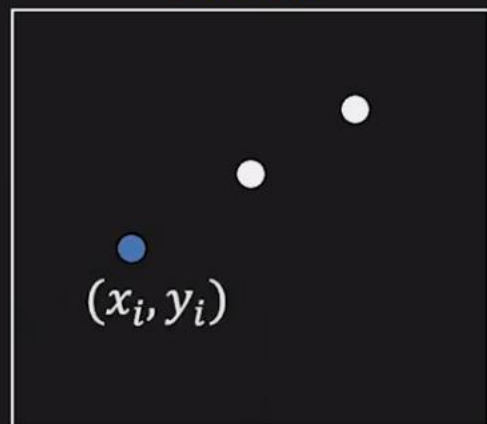
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$$A(m, c) = A(m, c) + 1$$

if (m, c) lies on the line: $c = -mx_i + y_i$

Image



$A(m, c)$

c	1	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0

Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create **accumulator array** $A(m, c)$

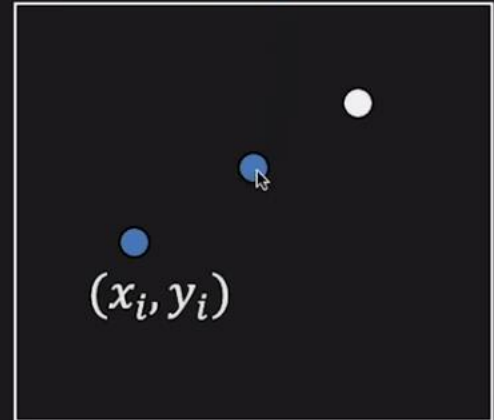
Step 3. Set $A(m, c) = 0$ for all (m, c)

Step 4. For each edge point (x_i, y_i) ,

$$A(m, c) = A(m, c) + 1$$

if (m, c) lies on the line: $c = -mx_i + y_i$

Image



$A(m, c)$

c					
	1	0	0	0	1
	0	1	0	1	0
	0	0	2	0	0
	0	1	0	1	0
	1	0	0	0	1

Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create **accumulator array** $A(m, c)$

Step 3. Set $A(m, c) = 0$ for all (m, c)

Step 4. For each edge point (x_i, y_i) ,

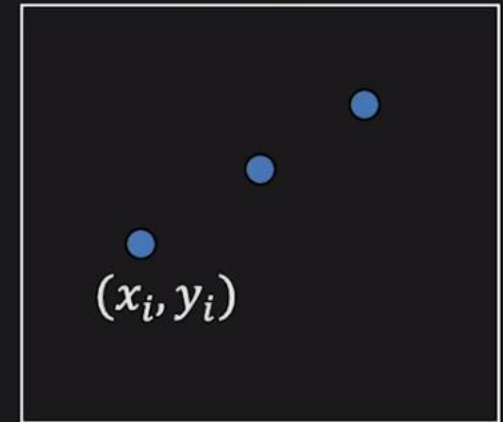
$$A(m, c) = A(m, c) + 1$$

if (m, c) lies on the line: $c = -mx_i + y_i$

Step 5. Find local maxima in $A(m, c)$

Draw the line using local maxima

Image

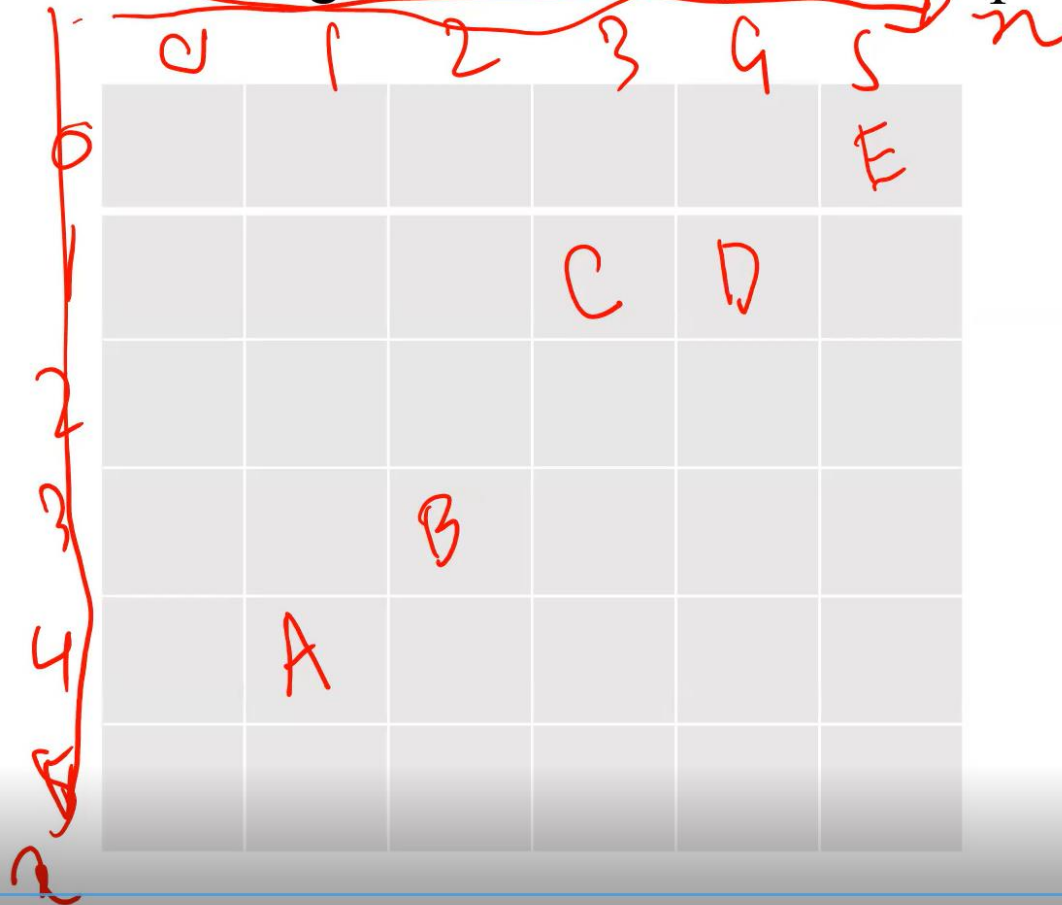


$A(m, c)$

c	1	0	0	0	1
	0	1	0	1	0
	1	1	3	1	1
	0	1	0	1	0
	1	0	0	0	1

Hough Transform Example

❖ Given a set of points $A(1,4)$, $B(2,3)$, $C(3,1)$, $D(4,1)$ and $E(5,0)$, use Hough transform to join these points.



$$y = mx + c$$

$$c = -mx + y$$

$$A(1,4) \quad c = -m + 4$$

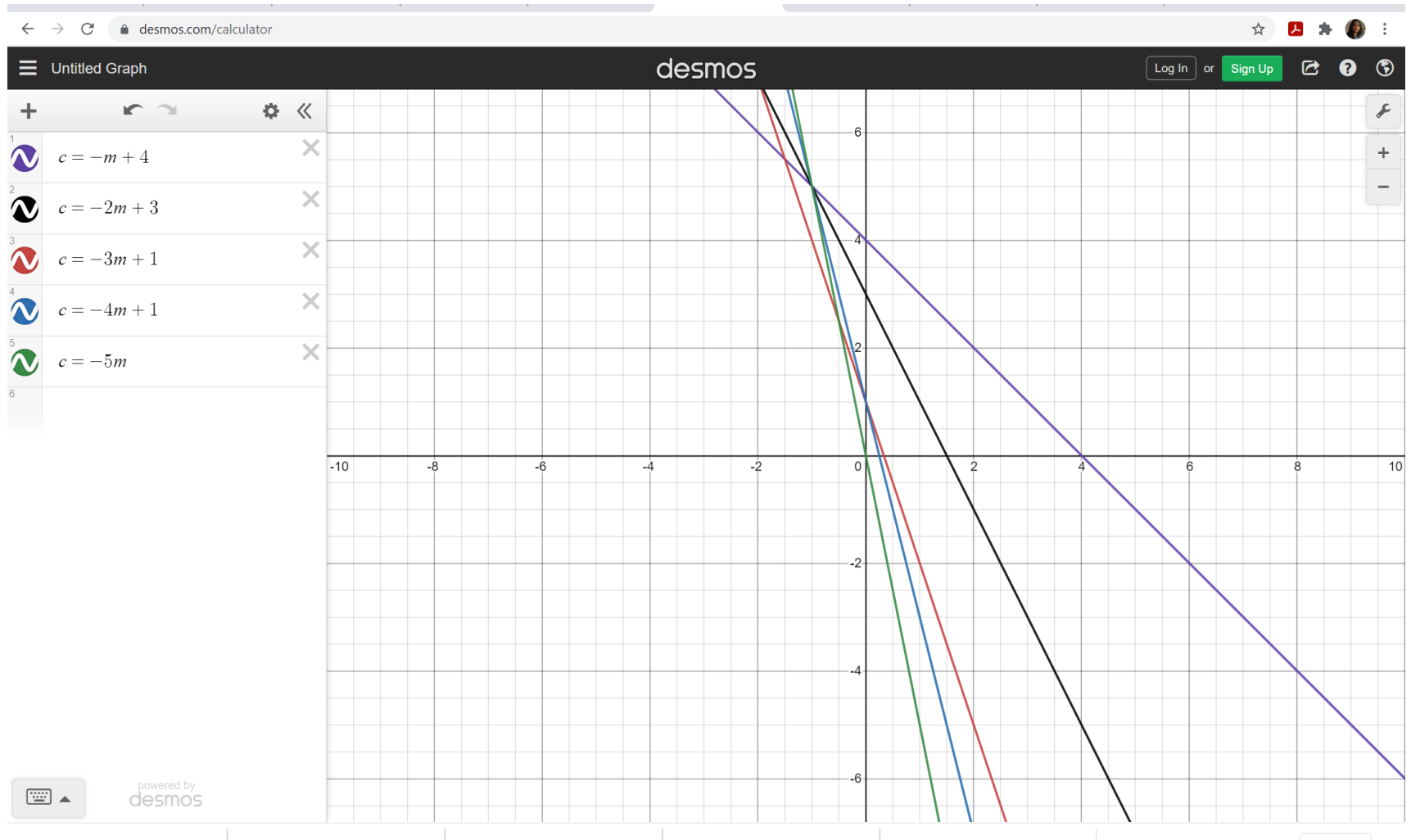
$$B(2,3) \quad c = -2m + 3$$

$$C(3,1) \quad c = -3m + 1$$

$$D(4,1) \quad c = -4m + 1$$

$$E(5,0) \quad c = -5m + 0$$

❖ Given a set of points A(1,4), B(2,3), C(3,1) D(4,1) and E(5,0), use Hough transform to join these points.



Hough Transform Example

- ❖ Given a set of points A(1,4), B(2,3), C(3,1) D(4,1) and E(5,0), use Hough transform to join these points.



<https://youtu.be/Lbu9SnoucW4>

Hough Transform Example

- ❖ Using Hough transform, show that the points $(1,2)$, $(2,2)$ and $(3,4)$ are collinear. Also find the equation of the line.

What is the problem in (m,b) parameter space / Hough space algorithm

- ❑ Limitations of this algorithm is that it does not work for vertical lines i.e. lines those are parallel to the y-axis.
- ❑ Because they have undefined or infinity slope.
- ❑ Therefore these lines must be converted into **polar coordinates**.

- Limitation of **previous algorithm** is that it does not work for vertical lines because they have infinity slope
- Therefore this **line** must be converted into **polar coordinates**

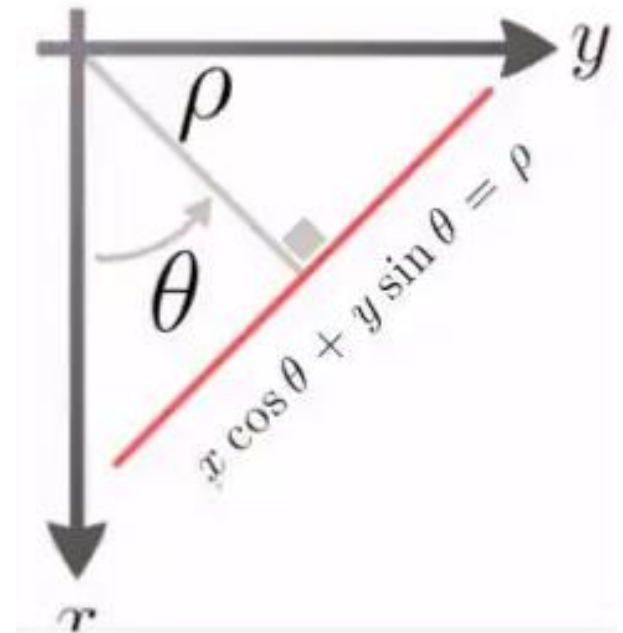
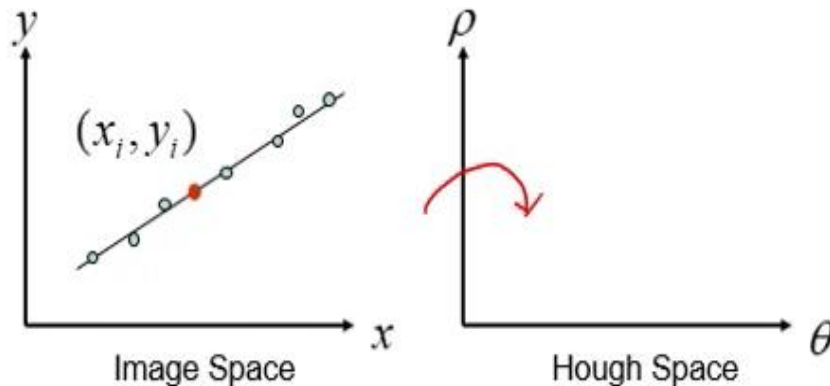
Equation of Line in polar:

$$\rho = x \cos \theta + y \sin \theta$$

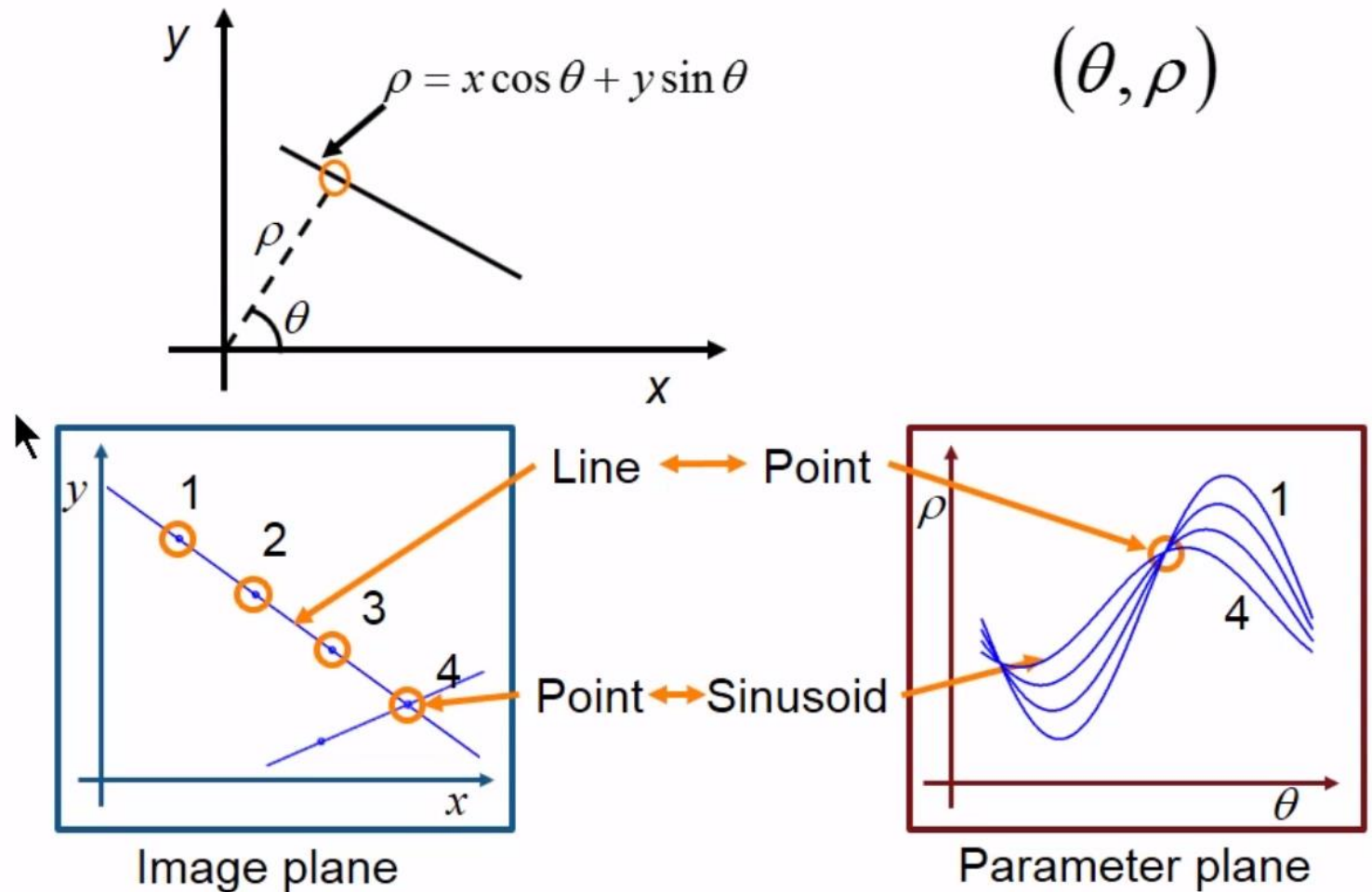
Where,

θ = angle between the line

ρ = diameter

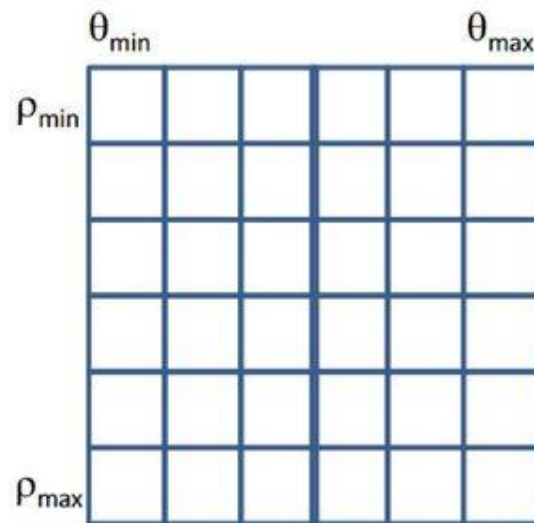
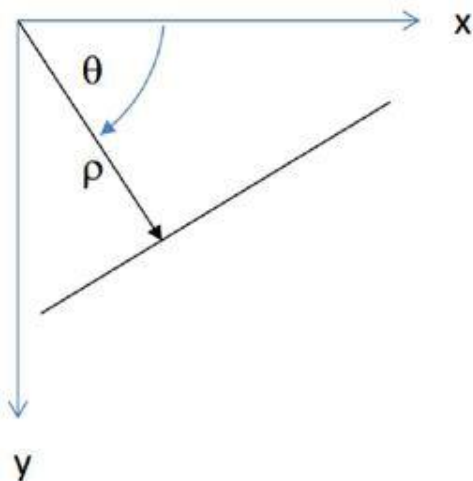


Standard Hough Transform



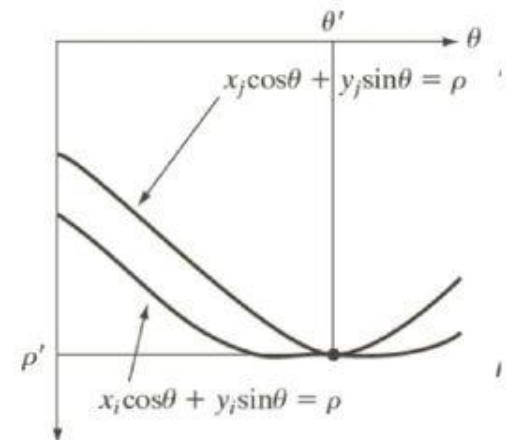
Polar Coordinate Representation of Line

- $\rho = x \cos \theta + y \sin \theta$
 - Avoids infinite slope
 - Constant resolution



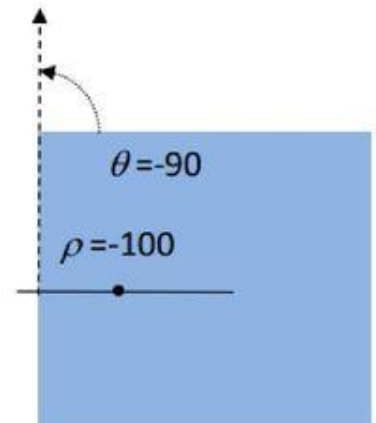
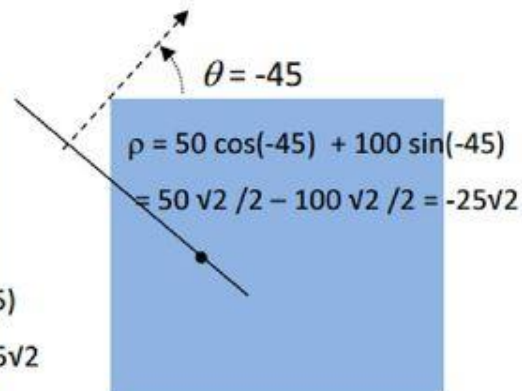
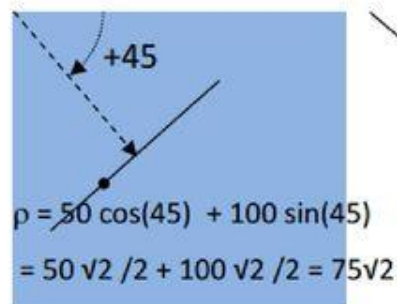
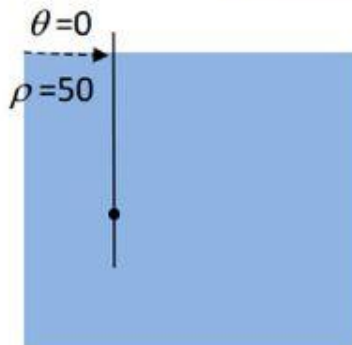
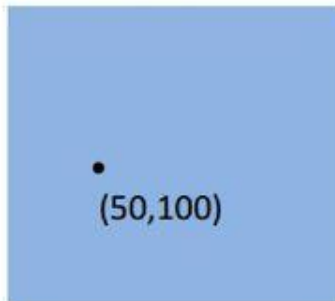
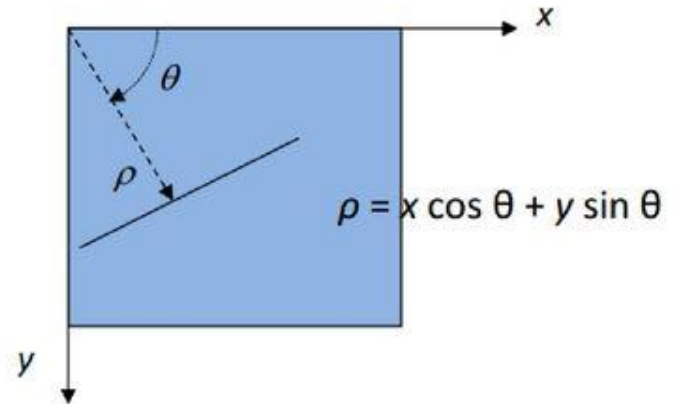
$A(\rho, \theta)$

The parameter space transform of a point is a sinusoidal curve



Hough Transform

- Angle, axis conventions
 - angle range is $-90^\circ \dots +89^\circ$
 - rho range is $-d_{\max} \dots +d_{\max}$
 - $d_{\max}$ is the largest possible distance
- Example of a point at $(x,y) = (50,100)$



Example

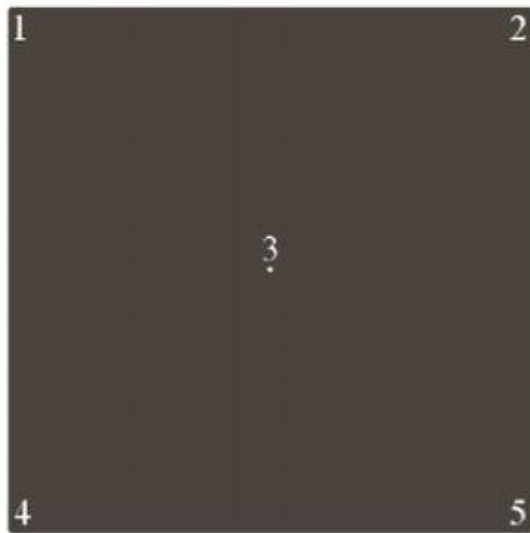
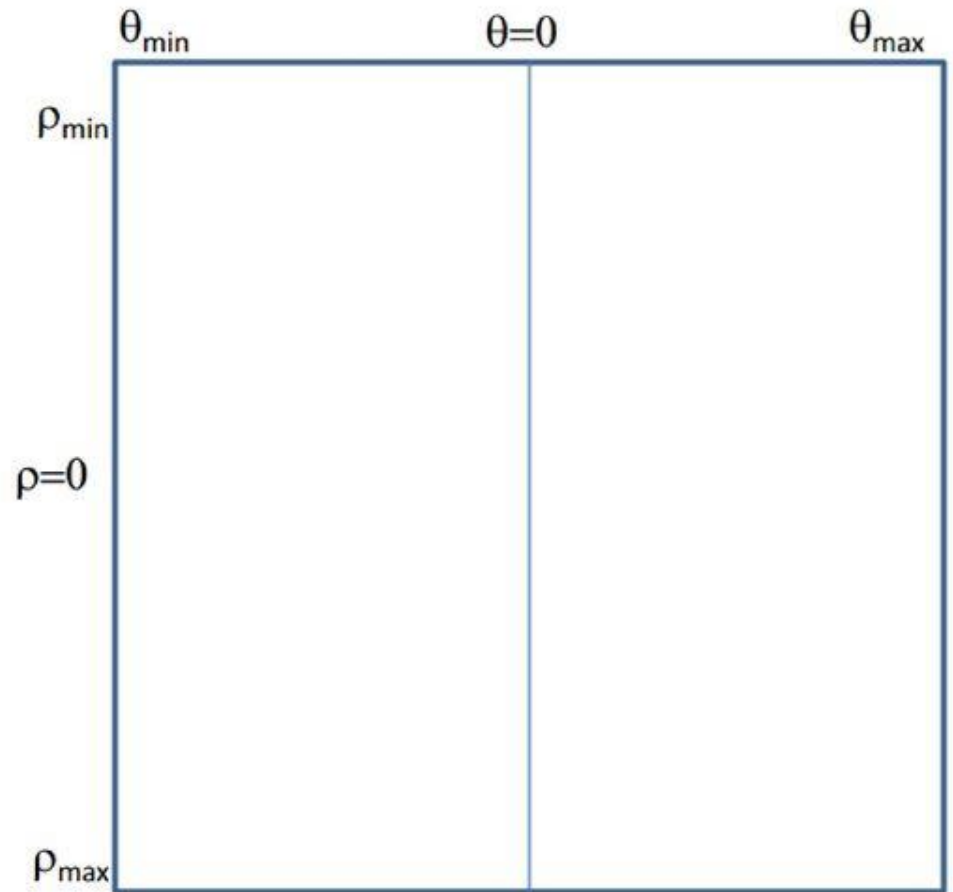


Image of size 100x100,
containing 5 points (at
the corners)



Example

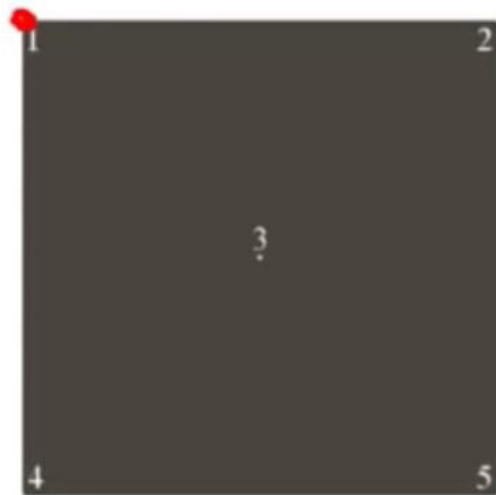
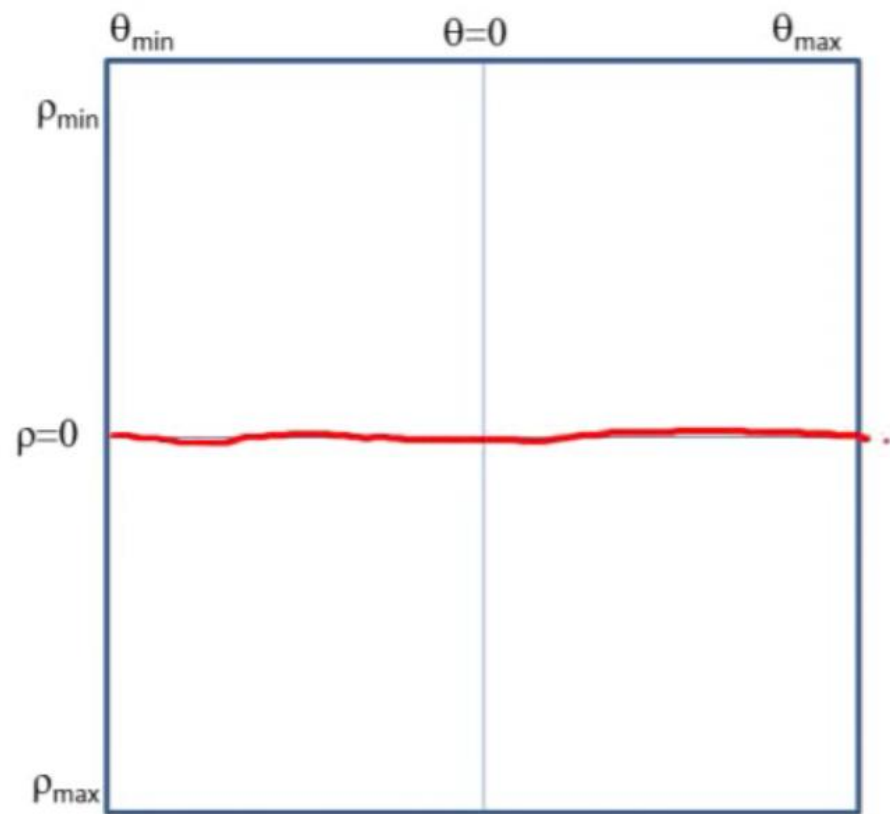


Image of size 100x100,
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Example

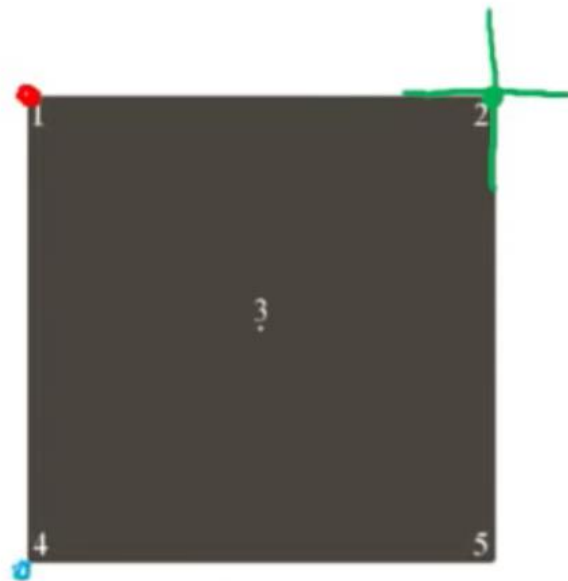
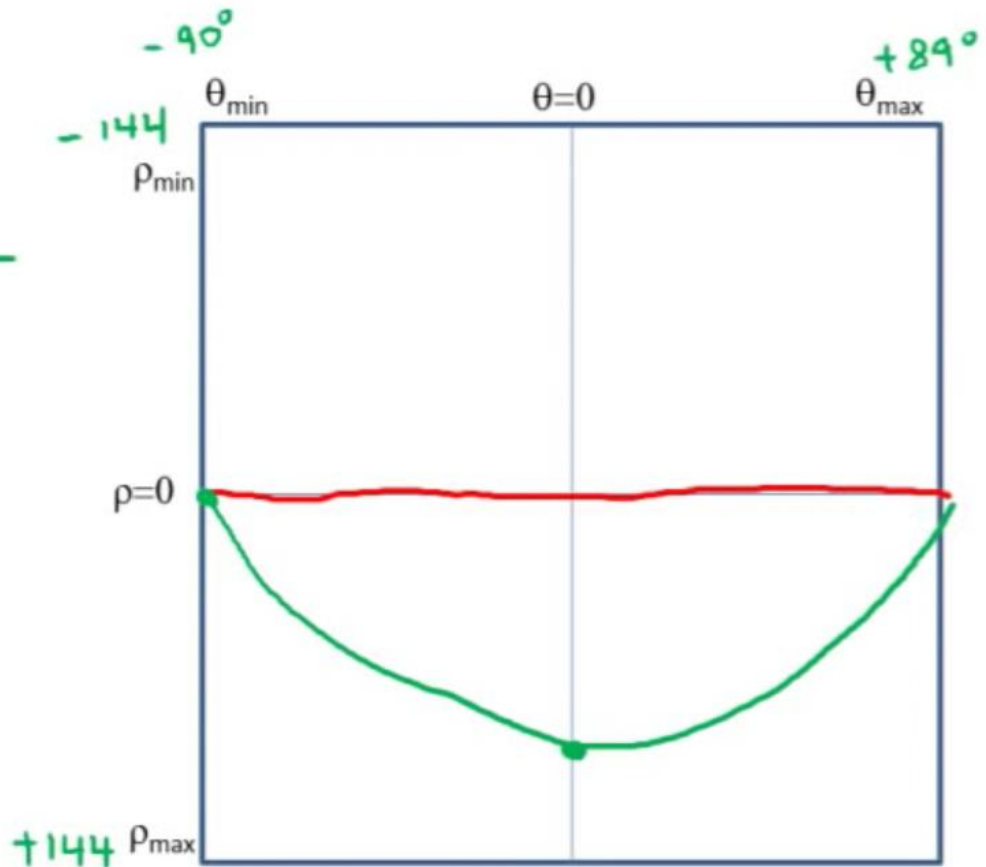


Image of size 100x100,
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the corners)



Example

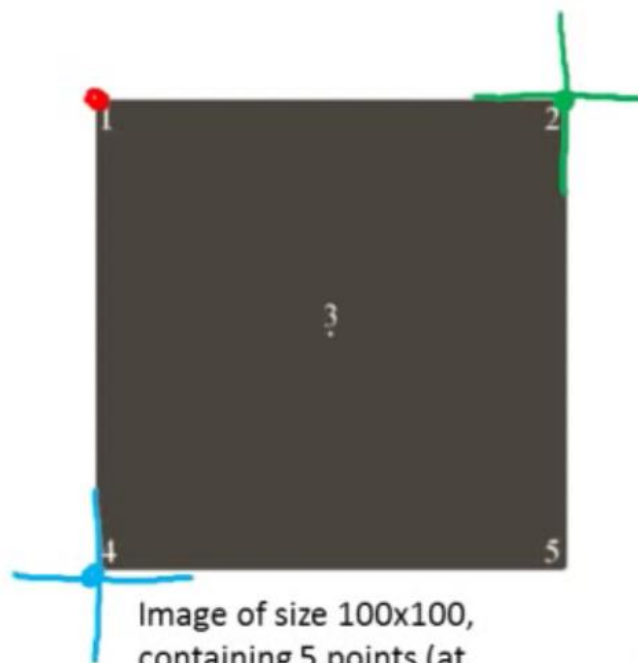
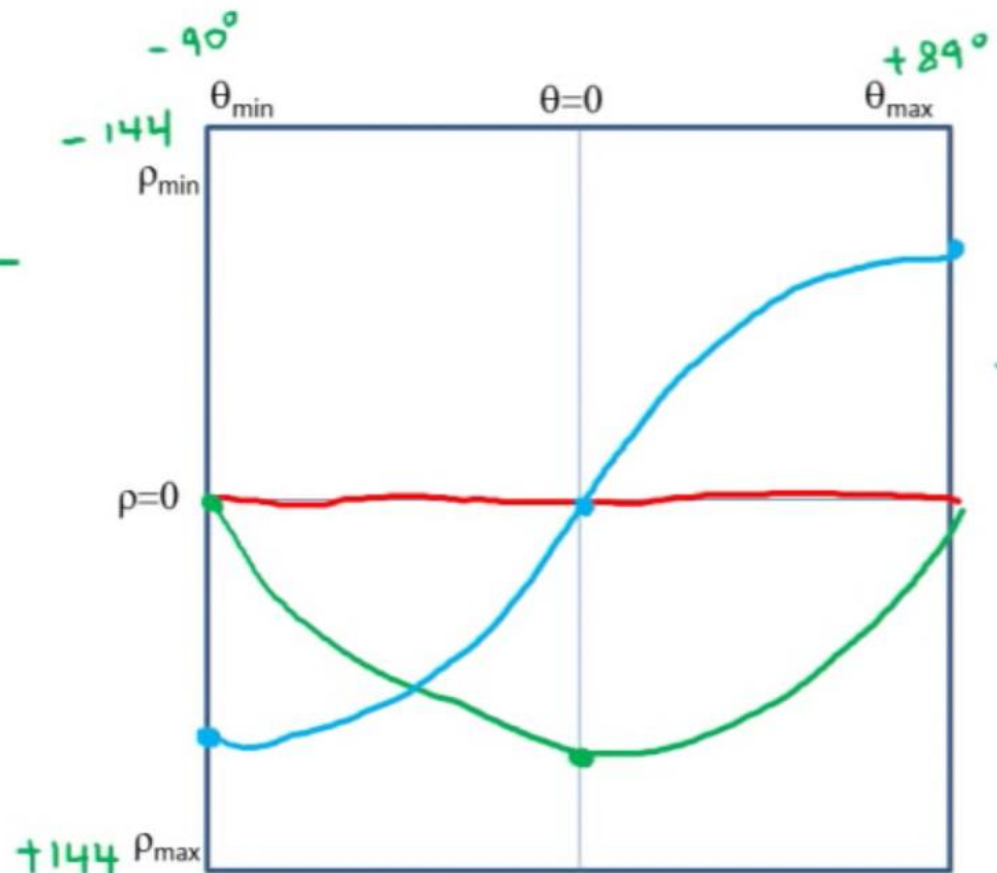
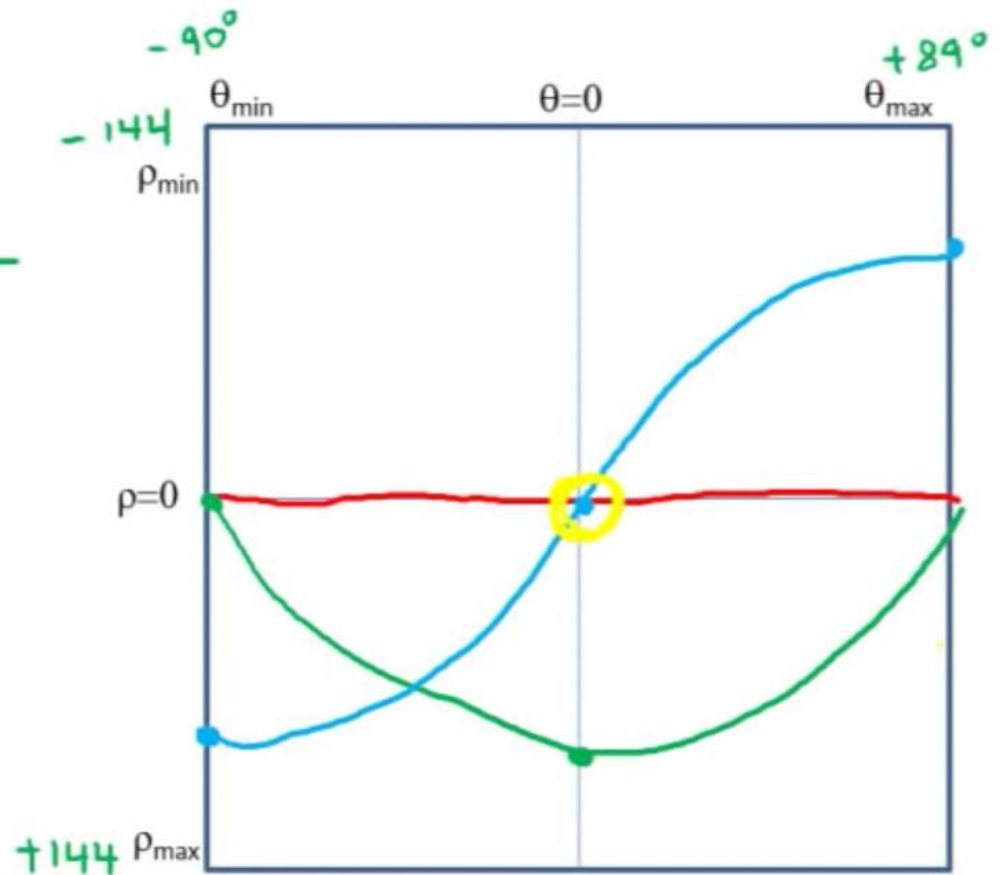
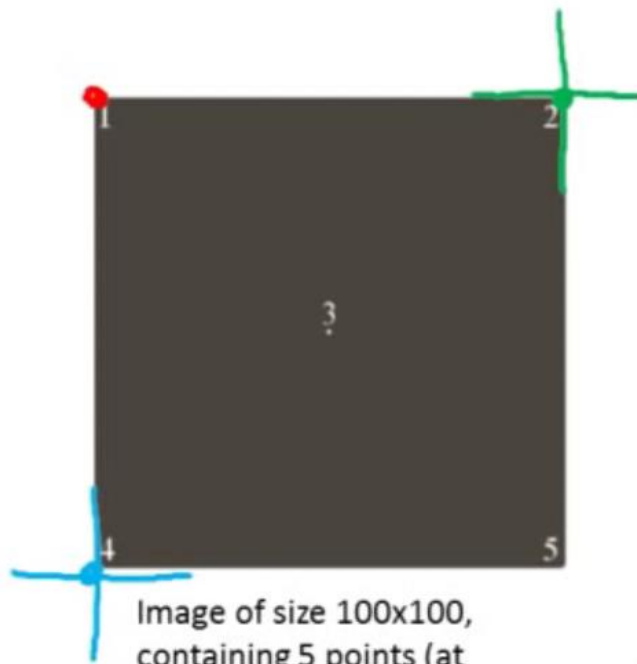


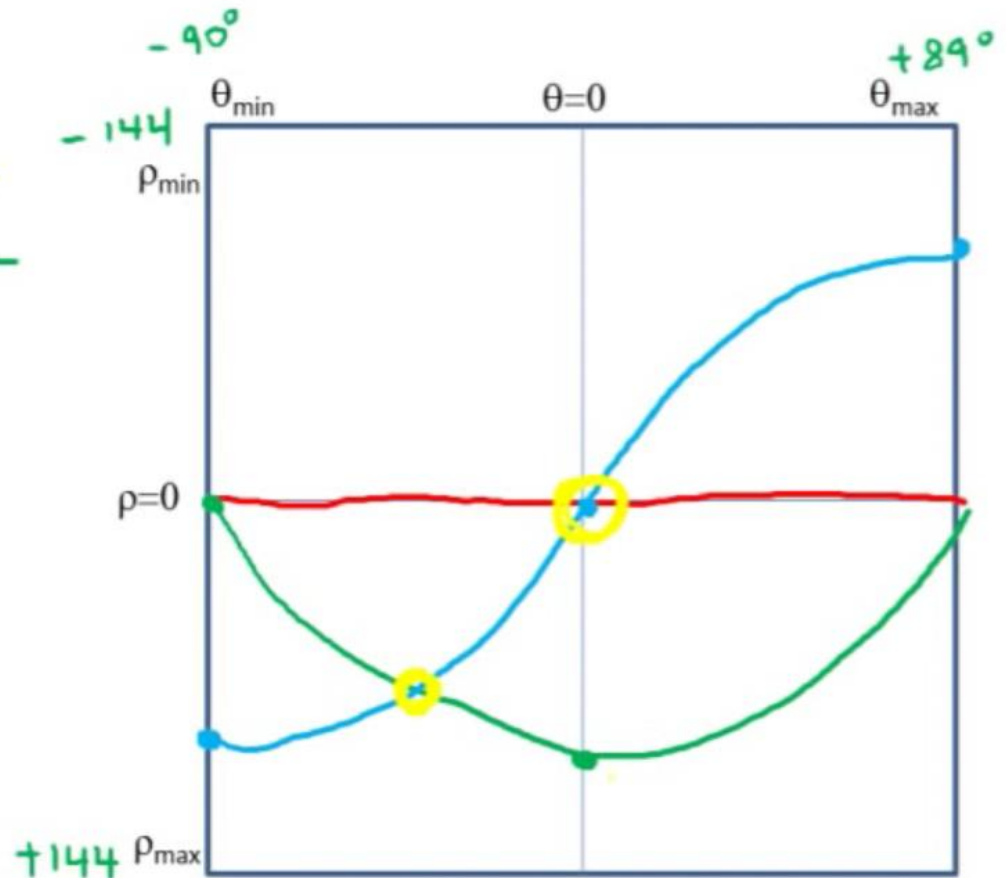
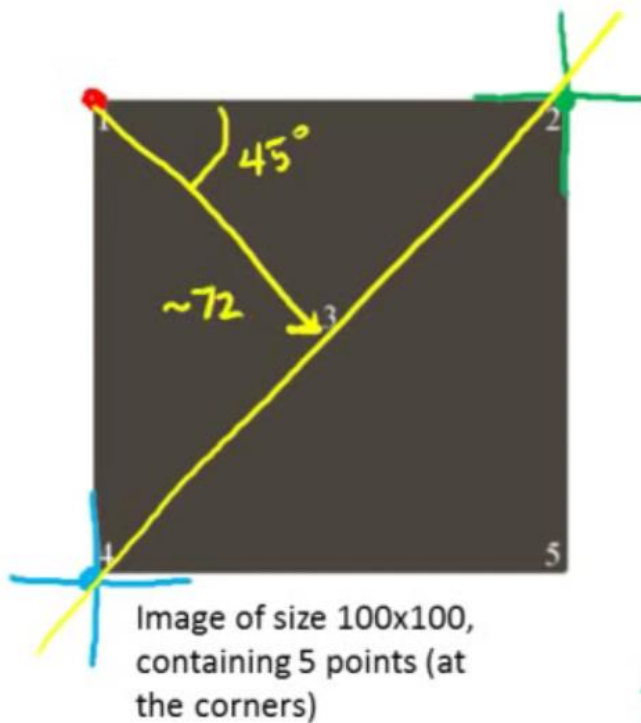
Image of size 100x100,
containing 5 points (at
the corners)



Example



Example



<https://youtu.be/o-n7NoLArcs>

Pseudo Code

```
for all x
  for all y
    if edge point at (x,y)
      for all theta
         $\rho = x \cdot \cos(\theta) + y \cdot \sin(\theta)$ 
        increment (add 1 to) the cell in H corresponding to (theta,rho)
      end
    end
  end
end
```

<https://youtu.be/NFtPH2REs28>

Digital Image Processing

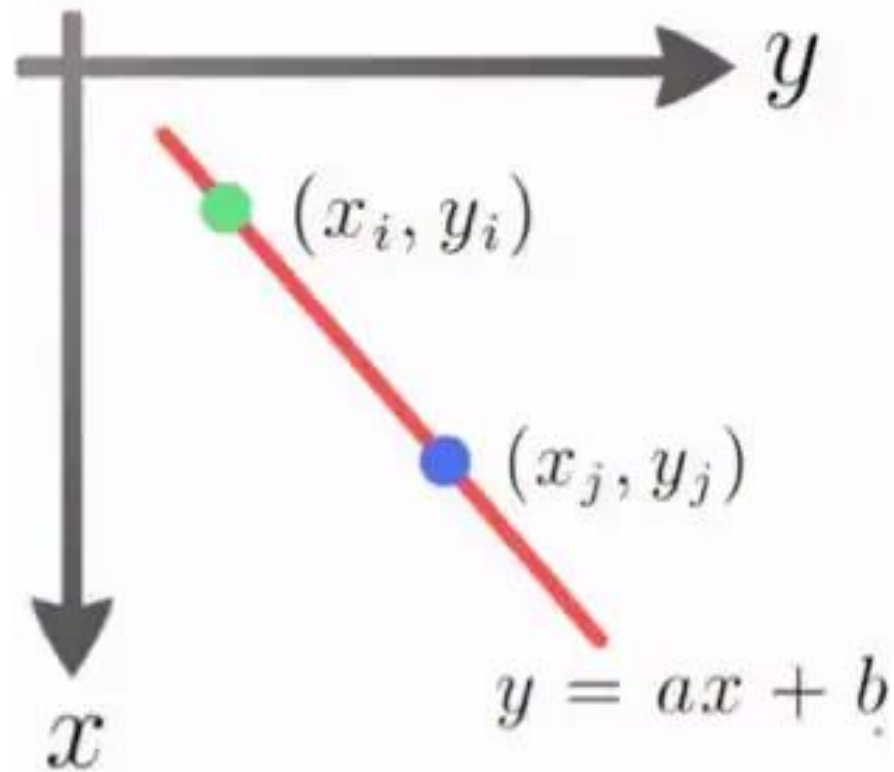
Video Lecture on Hough Transform and Shape Detection



Dr. Sapna Katiyar
Professor, ECE Department

Extra Slides

Hough Transform



Haugh Transform

- Steps:
- Consider one valid edge point (x_i, y_i) in xy-plane & the equation of line passing through it can be,

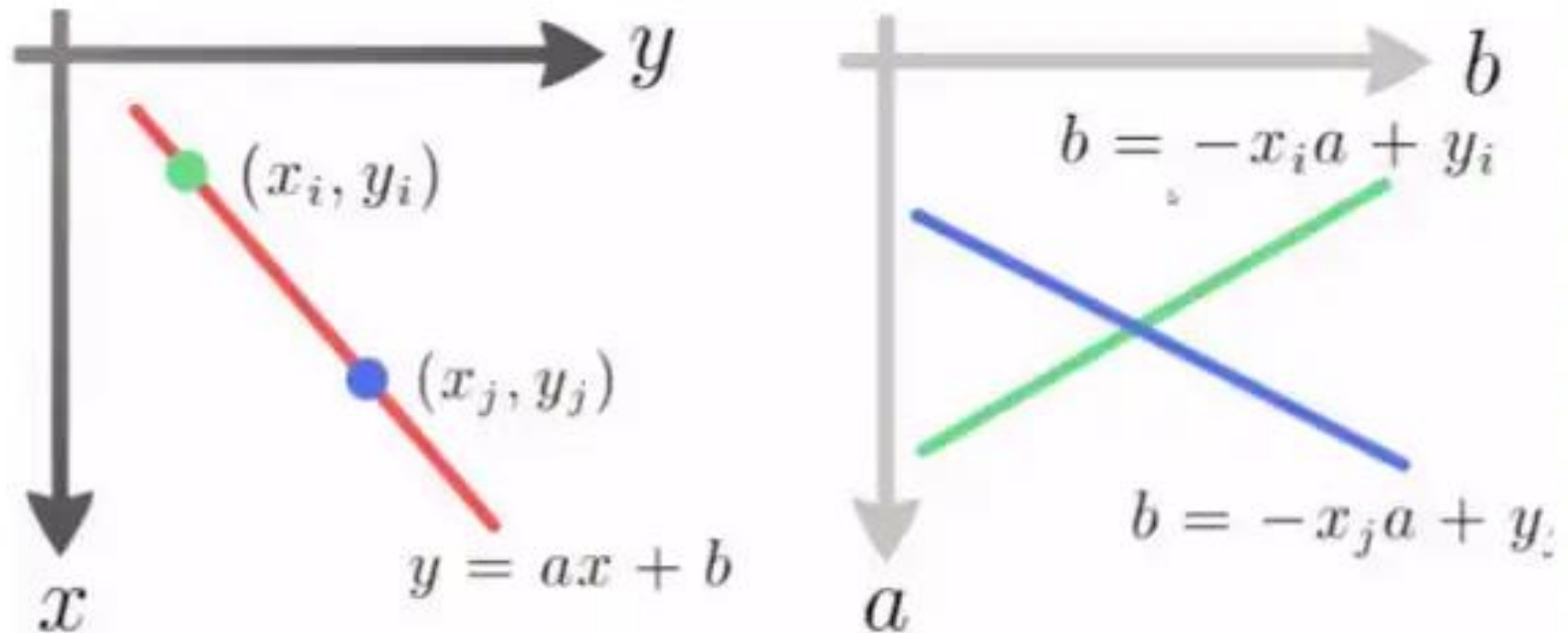
$$y_i = ax_i + b$$

- As it is a point, infinite lines will be passing through it given by above equation & different values of a & b .
- We can write this equation as,

$$b = -x_i a + y_i$$

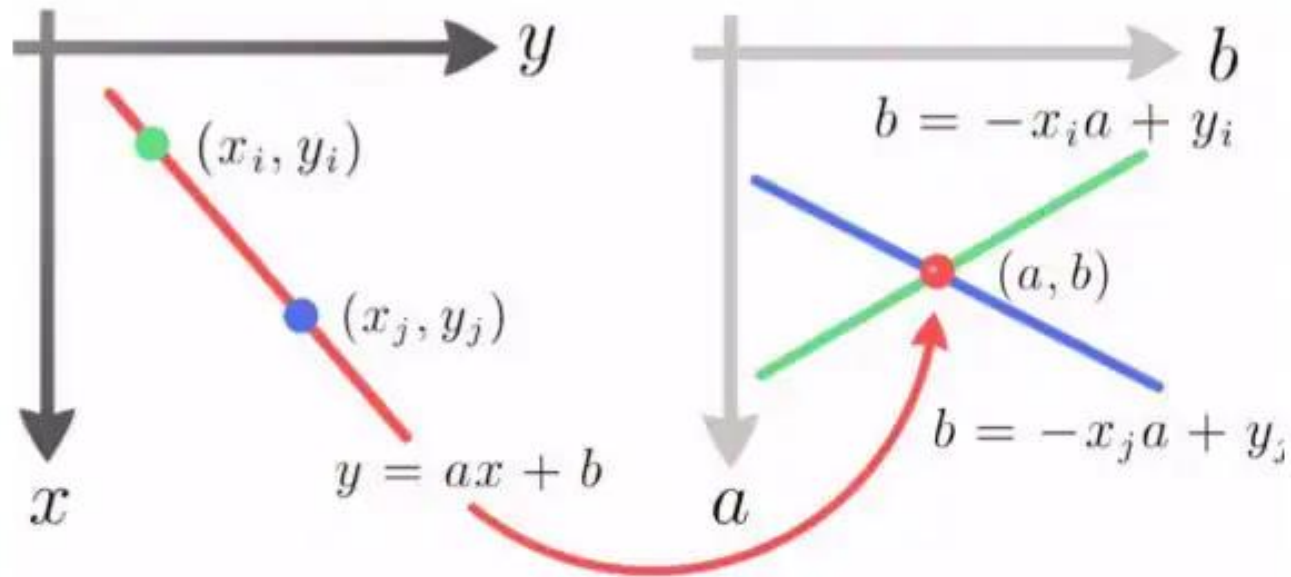
which gives us a line in ab -plane(parameter plane) passing through fixed pair (x_i, y_i) .

Hough Transform



- Next, we will consider 2nd valid edge point (x_j, y_j) and find out equation in parameter plane. It will be,

Hough Transform



- Next, we will consider 2nd valid edge point (x_j, y_j) and find out equation in parameter plane. It will be,

$$b = -x_j a + y_j$$

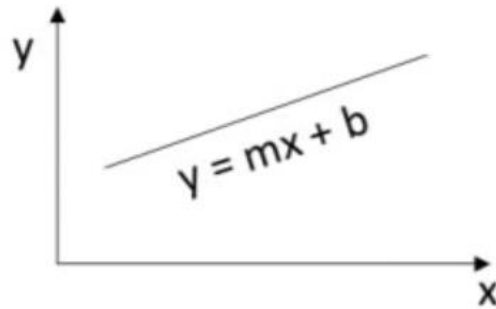
- If these 2 points lie on a straight line in xy -plane, then the two lines in parameter plane will intersect at point (a', b') where, a' is slope and b' is intercept of line passing through 2 points (x_i, y_i) and (x_j, y_j) in xy -plane.

Haugh Transform

- What is the drawback???
- Slope (a) will be infinite in case of vertical lines.
- Example:
 - If 2 valid points are (3,1) and (3,2)
 - Line eq in ab-plane will be,
 - $(3,1) \rightarrow b = -3a + 1$ & $(3,2) \rightarrow b = -3a + 2$
 - Here, slope of 2 lines is equal, hence they are parallel in ab-plane.
 - We can not find point of intersection which gives us slope and intercept i.e a' and b' of line passing through (3,1) and (3,2) in xy-plane.
- Solution???

Hough Transform (continued)

- A line has two parameters (m,b)



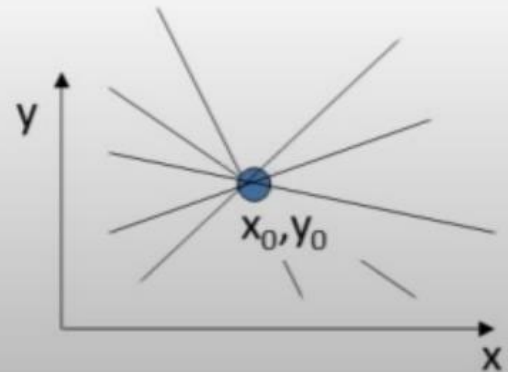
- Given a point (x_0, y_0) , the lines that could pass through this point are all (m,b) satisfying

$$y_0 = m x_0 + b$$

- Or

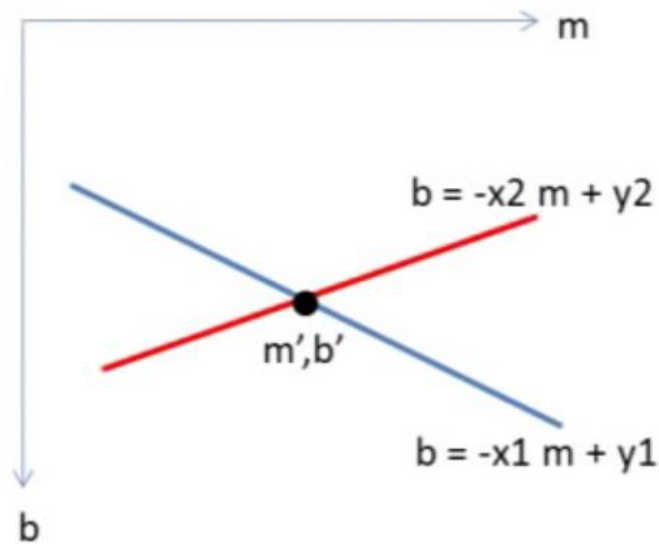
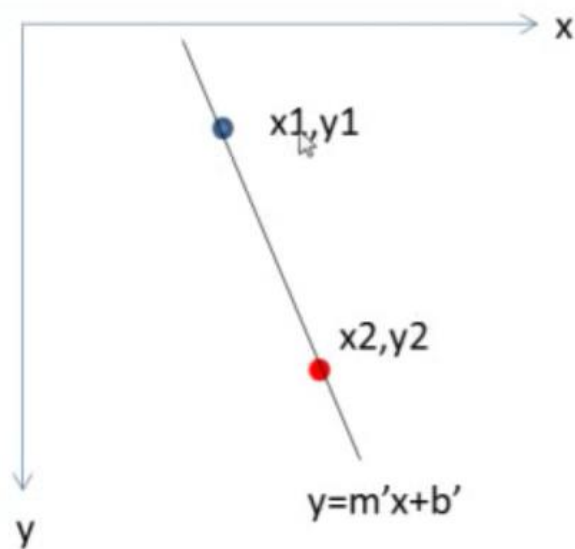
$$b = -x_0 m + y_0$$

The equation $b = -x_0 m + y_0$ is a line in (m,b) space

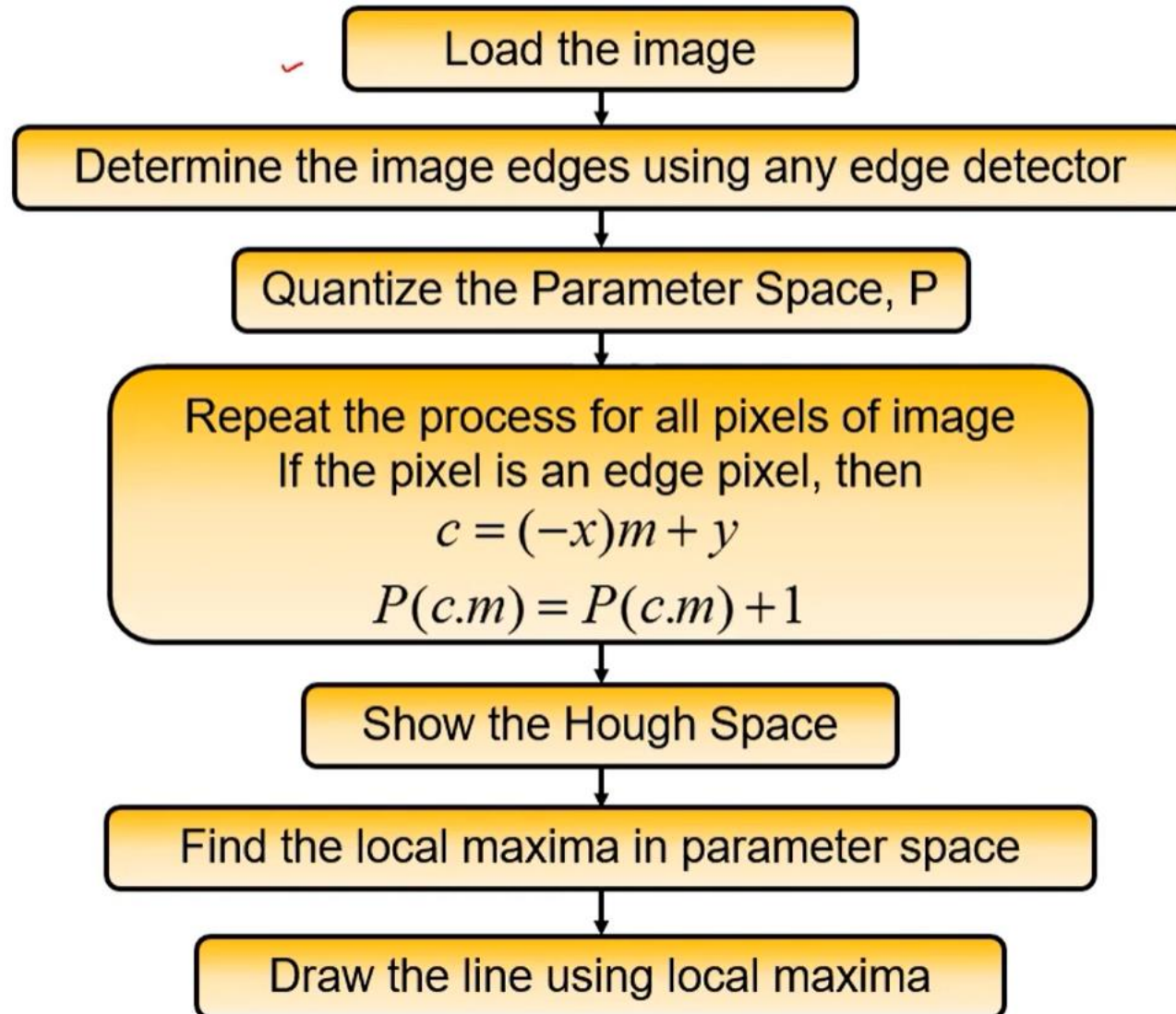


Hough Transform (continued)

- All points on a line in image space, yield lines in parameter space which intersect at a common point
 - This point is the (m,b) of the line in image space

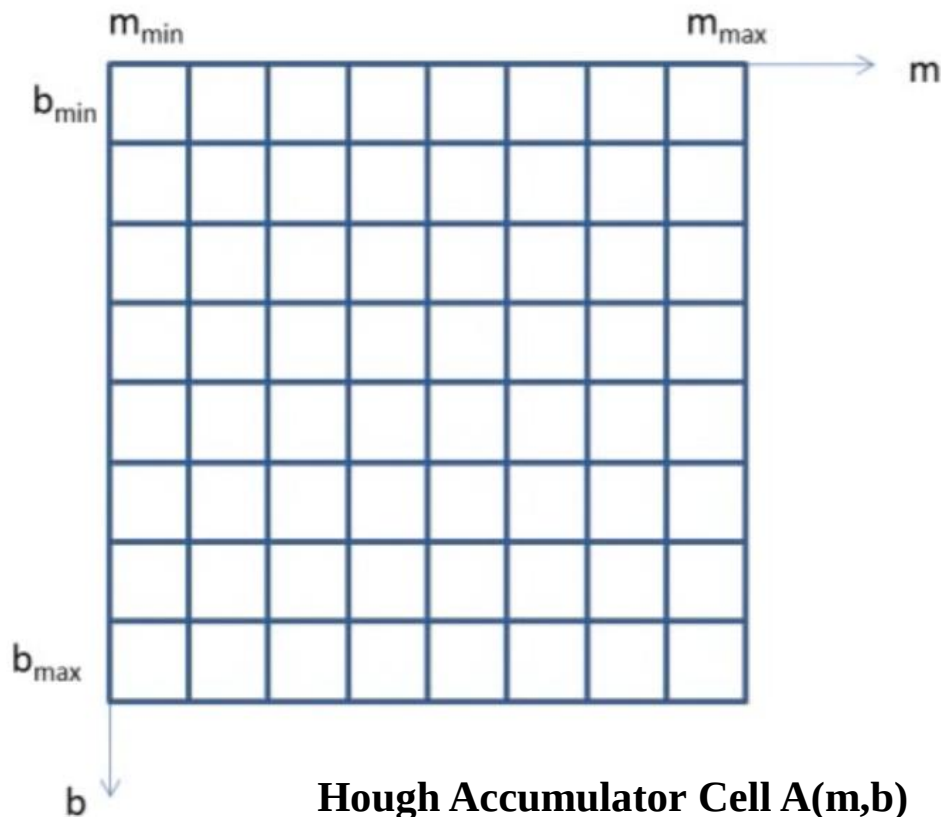


Hough Transform Algorithm



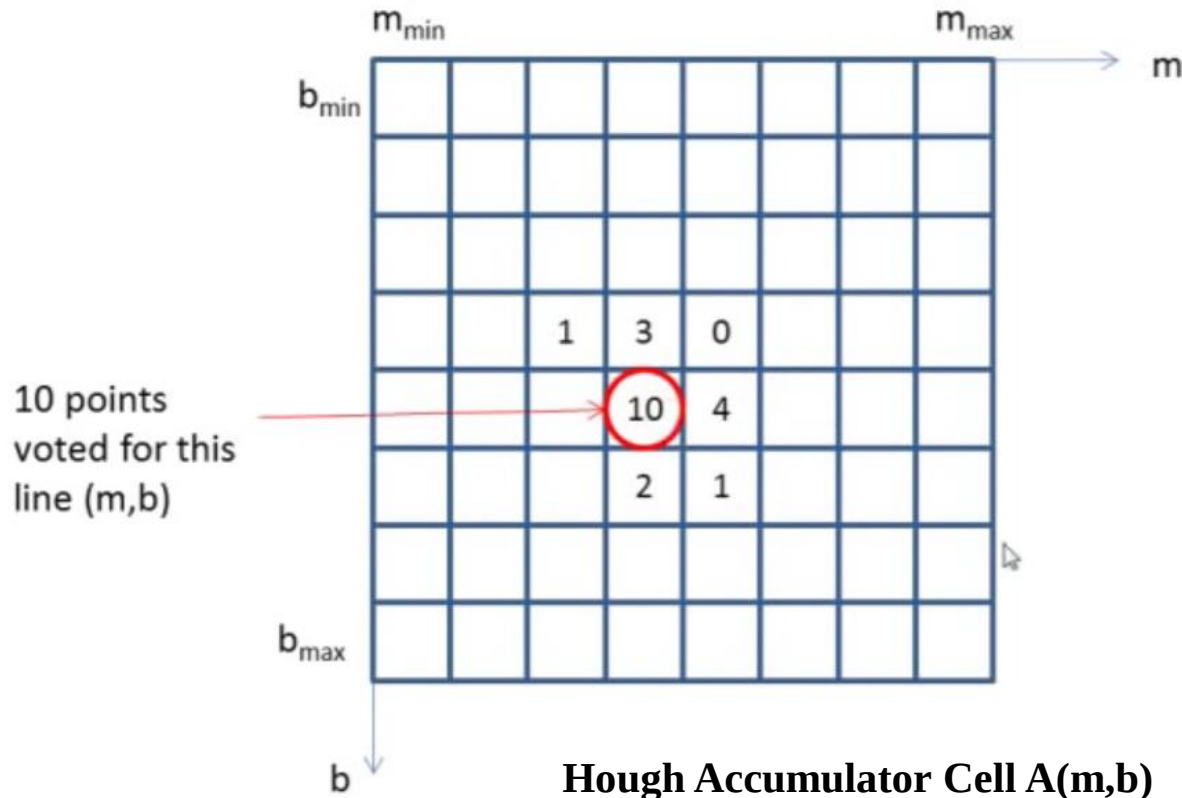
Hough Transform Algorithm

- Initialize an accumulator array $A(m,b)$ to zero



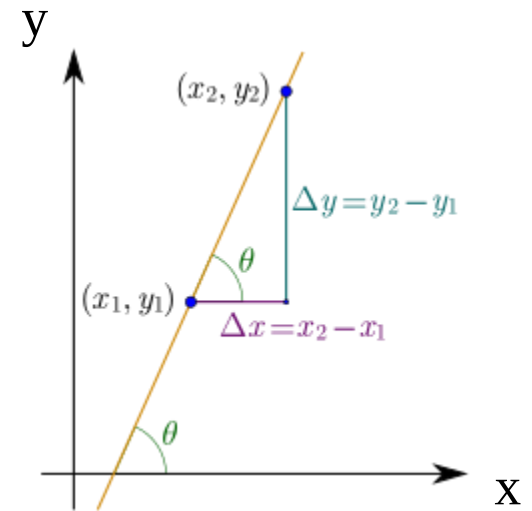
Hough Transform Algorithm

- Initialize an accumulator array $A(m,b)$ to zero
- For each edge element (x,y) , increment all cells that satisfy $b = -x m + y$
- Local maxima in $A(m,b)$ correspond to lines



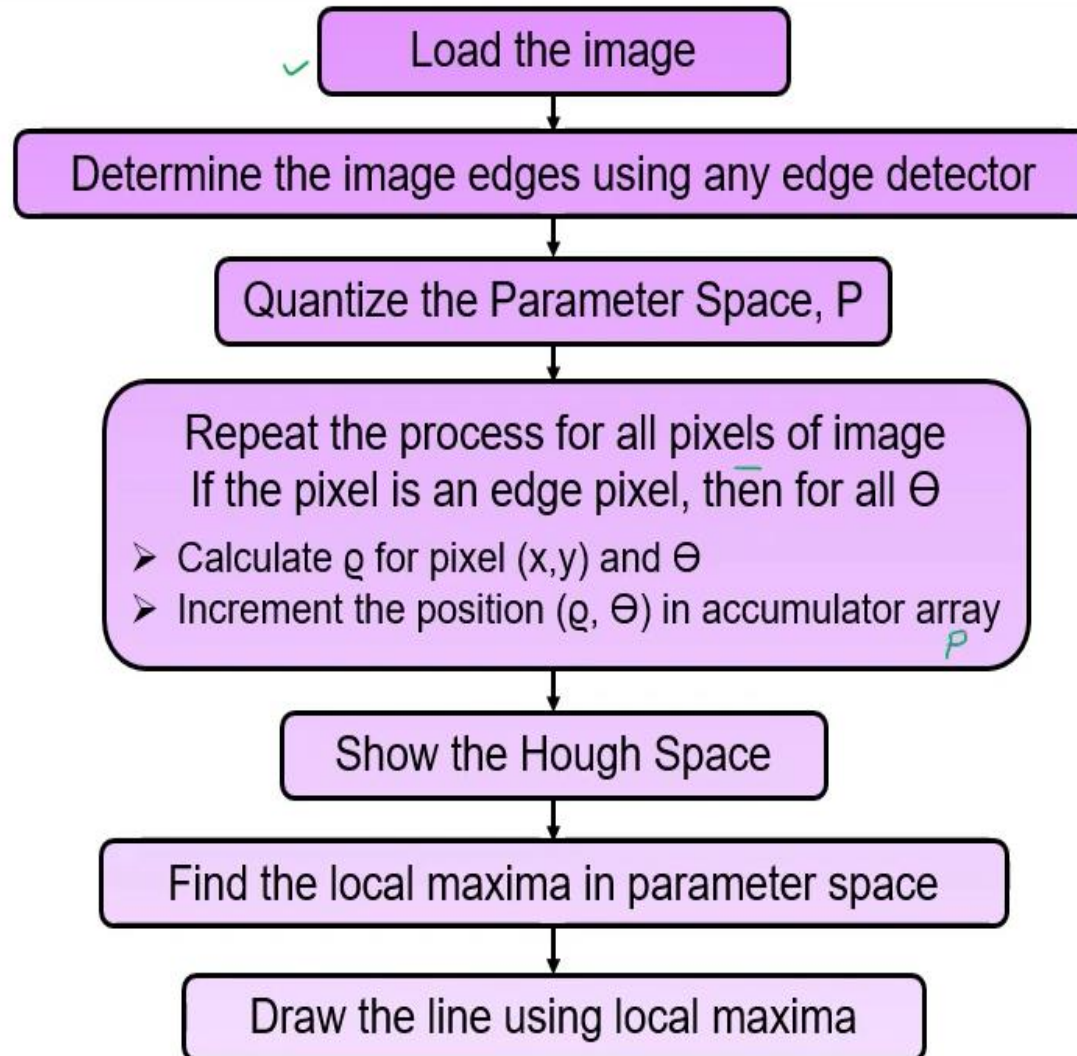
What is the slope of a line parallel to the y-axis?

- ❑ Parallel lines have the same slope.
- ❑ Vertical lines have an undefined slope.
- ❑ The y-axis is a vertical.
- ❑ A line that is parallel to the y-axis also has to be vertical.
- ❑ **The slope of a line parallel to the y-axis has a slope that is undefined.**



$$\text{Slope} = m = \tan\theta$$

Hough Transform Algorithm in polar space



Better Parameterization

NOTE: $-\infty \leq m \leq \infty$

Large Accumulator

More memory and computations

Improvement: (Finite Accumulator Array Size)

Line equation: $\rho = -x \cos \theta + y \sin \theta$

Here $0 \leq \theta \leq 2\pi$

$$0 \leq \rho \leq \rho_{\max}$$

Given points (x_i, y_i) find (ρ, θ)

Hough Space Sinusoid

